Client-side Technologies

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iTi

Day 7

JavaScript Fundamentals cont.

Form

- By using the form you have at your disposal; information about the elements in a form and their values.
- A separate instance of the form object is created for each form in a document.
- Objects within a form can be referred to by a numeric index or be referred to by name.
- **Object Model Reference:**

  [window.]document.formname

  [window.]document.forms[i]

  [window.]document.forms[“formNAME”]

  [window.]document.forms[“formID”]

Form

```
<form
    [name="formName"]
    [target="frameName or windowName"]
    [onsubmit="handlerText Or Function"]
    [onreset="handlerText Or Function"]
>
</form>
```

Properties

Events

Form Properties

Property	Description
elements[ ]	An array containing all of the elements of the form. Use it to loop through form easily.
length	The number of elements in the form.
name	The name of the form.
id	The id of the form.
target	The name of the target frame or window form is to be submitted to.

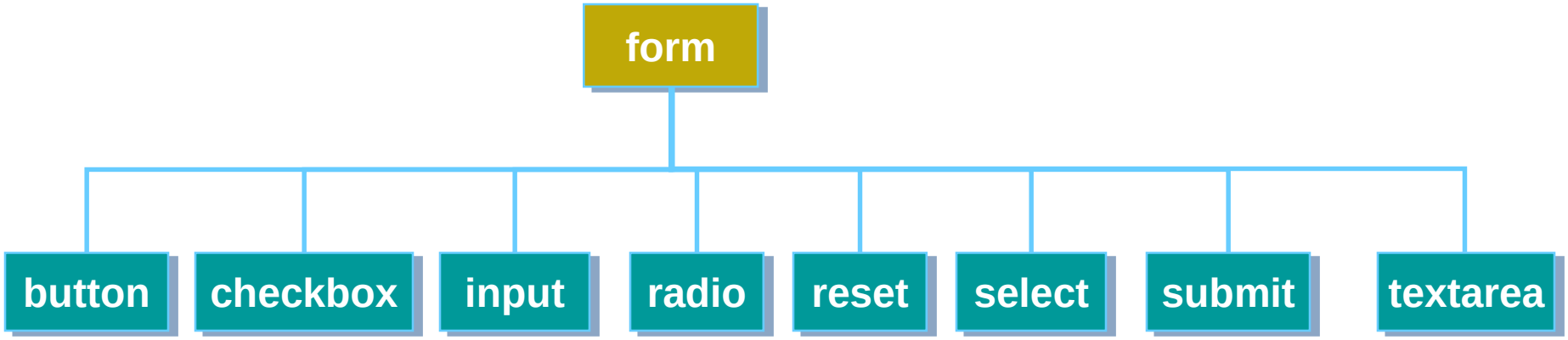
Form Methods

Method	Description
reset()	Resets the form. Clicking the reset button clears all contents that the user has made..
submit()	Submits a form. Clicking the submit button submits the content of the form to the server

Form Event Handler

Event	Description
onreset	Code is executed when the form is reset (by clicking on "reset" button)
onsubmit	Code is executed when form is submitted

Form


form
button
checkbox
input
radio
reset
select
submit
textarea

Text

```
<input type="text"
      id="id"
      value="string"
     maxlength="n"
     size="x"
/>
```

Text Event Handler

Event	Event Handler	Description
focus	onfocus	Fires when the field gains focus when the user tabs into or clicks inside the control.
blur	onblur	Fires when the field loses focus when the user tabs from or clicks outside the control.
change	onchange	Fires after changing the field value and losing being focused.
input	oninput	Fires every time the field value changes.
select	onselect	Fires when the content of the field being selected.

Text Methods

Method	Description
blur( )	Removes focus from the field.
focus( )	Assigns focus to the field; places the cursor in the control.
select( )	Selects, or highlights, the content of the control.

Example!

Drop-down lists

```
<select
    id="id"
    multiple
    size="n"
>
  <option value="string" selected > label </option>
  <option value="string2" > label2 </option>
  ...
</select>
```

Drop-down lists Properties

Property	Description
length	The number of options in the list.
selectedIndex	The index number, beginning with 0, of the selected option.
options[ ]	An array of the options in the list. Used to reference properties associated with the options; e.g., options[1].value or options[2].text.
selected	A true or false value indicating whether an option is chosen.
value	The value associated with an option
text|label	The text label associated with an option.

Drop-down lists Event Handler

Event Handler	Description
onfocus	The control gains focus.
onblur	The control loses focus.
onchange	A different option from the one currently displayed is chosen.

Example!

Radio Button

```
<input type="radio"
      id="id"
       name="name"
       value="string"
       checked
/>
```

Radio Button Properties

Property	Description
length	The number of radio buttons with the same name.
checked	A true or false value indicating whether a button is checked.
value	The value attribute coded for a button. A checked button with no assigned value is given a value of "on".

Radio Button Event Handler

Event Handler	Description
onfocus	The control gains focus.
onblur	The control loses focus.
onclick	The button is clicked.

Example!

Checkbox

```
<input type="checkbox"
    id="id"
    name="name"
    value="string"
    checked
/>
```

Button

```
<input type="button“
    id="id“
    value="string"
/>
```

Button Event Handler

Event Handler	Description
onclick	The mouse is clicked and released with the cursor positioned over the button.
ondblclick	The mouse is double-clicked with the cursor positioned over the button.
onmousedown	The mouse is pressed down with the cursor positioned over the button.
onmouseout	The mouse cursor is moved off the button.
onmouseover	The mouse cursor is moved on top of the button.
onmouseup	The mouse button is released with the cursor positioned over the button.

Reminder DOM References

- Use **this** keyword is used to refer to the current object.
  - e.g. the calling object in a method.

- Self reference to the object is used :

```
<input   type="text"
                onfocus = "this.value='You are in!'"/>
```

- Passing current Object as a function parameter:

```
function myFunction(myObject) {
        myObject.value = "In the function!!"
}
<input type="text" onclick="myFunction(this)"/>
```

Events & Event Handlers

Events

- We have the ability to create dynamic web pages by using *events*.
- Events are actions that respond to user's specific actions.
- Events are controlled in JavaScript using event handlers that indicate what actions the browser takes in response to an event.
- Examples for different events:
  - A mouse click
  - A web page loading
  - Taking mouse over an element
  - Submitting an HTML form
  - A keystroke on your keyboard

Events

- Event handlers are created as attributes added to the HTML tags in which the event is triggered. (first way of binding an event handler)

- An Event handler adopts the event name and appends the word "*on*" in front of it.

  < tag onevent = "JavaScript commands;">

- Thus the "click" event becomes the *onclick* event handler.

Mouse Events

Event handler	Description
onmousedown	when pressing any of the mouse buttons.
onmousemove	when the user moves the mouse pointer within an element.
onmouseout	when moving the mouse pointer out of an element.
onmouseup	when the user releases any mouse button pressed
onmouseover	when the user moves the mouse pointer over an element.
onclick	when clicking the left mouse button on an element.
ondblclick	when Double-clicking the left mouse button on an element.
ondragstart	When the user has begun to select an element

Keyboard Events

Event handler	Description
onkeydown	When User presses a key
onkeypress	When User presses a key other than Modifiers (ctrl, shft, ..etc.)
onkeyup	When User releases the pressed a key

Other Events

Event handler	Description
onabort	The User interrupted the transfer of an image
onblur	when loosing focus
onfocus	when setting focus
onchange	when the element has lost the focus and the content of the element has changed
onload	a document or other external element has completed downloading all the data into the browser
onunload	a document is about to be unloaded from the window
onerror	When an error has occurred in a script.
onmove	when moving the browser window

Other Events

Event handler	Description
onreset	When the user clicks the form reset button
onsubmit	When the user clicks the form submit button
onscroll	When the user adjusts an element's scrollbar
onresize	When the user resizes a browser window
onhelp	When the user presses the F1 key
onselect	When selecting text in an input or a textarea element
onstart	When a marquee element loop begins
onfinish	When a marquee object finishes looping
onselectstart	When the user is beginning to select an element

Binding Events

- Binding Event Handlers to Elements can be:
    1. Event handlers as tag attribute
    2. Event handlers as object property

Event handlers as tag attribute

```
<input type=button value="click me" name=b1
   onclick="alert('you have made a click')">
```

OR

```
   <script>
   function showmsg(){
         alert("you have made a click")
   }
   </script>

<input type=button value="click me"
   onclick="showmsg()" />
```

Example!

Event handlers as object property

```
<body>
    <form>
      <input type=button name='b1' value="Click ME" />
    </form>
</body>
```

```
<script>
function showAlert (){
    alert("you have clicked me")
}
document.forms[0].b1.onclick=showAlert
</script>
```

Note: there are no parentheses

Event handlers as object property

```
<body>
    <form>
        <input type=button name='b1' value="Click ME" />
    </form>
</body>
```

```
<script>
document.forms[0].b1.onclick=function showAlert (){
                                    alert("you have clicked me")
                                }
</script>
```

Event handlers return value

```
<a href="1.htm" onclick="myFunc(); return false">
```

This will make the browser ignore the action of href

- Another way that can also make the browser ignore the action of href is:

```
<a href="javascript:void(0)" onclick="alert('hi')" >
    click me
</a>
```

To avoid refresh action
if href is empty

Example!

void Operator

- void Operator
  - A unary operator used to explicitly return undefined.
  - It can be used as shown

```
void expression;
void(expression);
```

- Example:

```
var  val= void "javascript";
typeof val;       //undefined
```

Event Object

Event Object

- The event object gives information about an event that has occurred.
- When an event occurs, an **event** object is initialized automatically and passed to the event handlers.
- We can create event object via its constructor

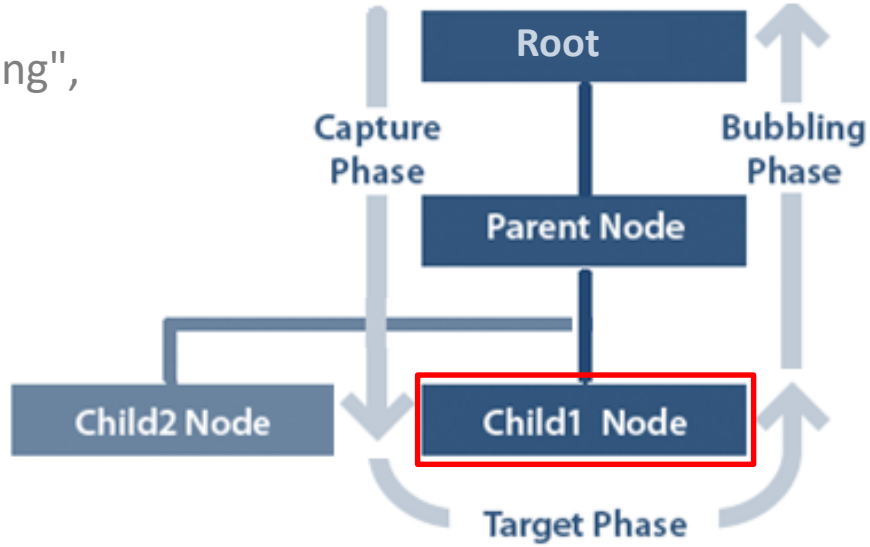
  var evt= new Event()

- The Event object represents the state of an event, such as the element in which the event occurred, the state of the keyboard keys, the location of the mouse, and the state of the mouse buttons.
- **Object Model reference:**

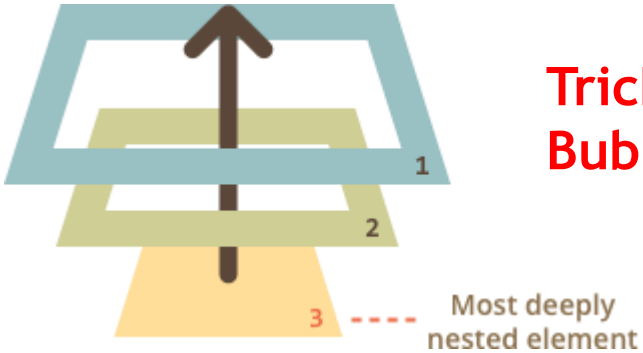
  [window.]event

Event Object

- Events always propagate from the root
- When an event occurs, it is dispatched to the target element first.
- 2 ways for objects to handle fired events
  - Event Capture (Phase1)
    - Capturing is also called "trickling",
    - Event goes down,
  - Event Bubbling (Phase3)
    - Event goes up



Event Object



Trickle down,
Bubble up

- If the event propagates up, then it will be dispatched to the ancestor elements of the target element in the DOM hierarchy.

- The propagation can be stopped with the stopPropagation() method and/or the cancelBubble property.

Event Object Properties

Event Object Property	Description
srcElement	The element that fired the event
target	
currentTarget	identifies the current target for the event, as the event traverses the DOM
type	String representing the type of event.
clientX (layerX)	Mouse pointer X coordinate at the time of the event occurs relative to upper-left corner of the window.
clientY (layerY)	Mouse pointer Y coordinate at the time of the event occurs relative to upper-left corner of the window.
offsetX	Mouse pointer X coordinate relative to element that fired the event.
offsetY	Mouse pointer Y coordinate relative to element that fired the event.

Event Object Properties

Event Object Property	Description
altKey	True if the alt key was also pressed
ctrlKey	True if the alt key was also pressed
shiftKey	True if the alt key was also pressed
button	Any mouse buttons that are pressed
keyCode(deprecated)	Returns UniCode value of key pressed use code property instead
which(deprecated)	
key	Represents the value of the key pressed (e.g. n)
code	Represents a physical key on the keyboard (e.g. keyN)

For alt,ctrl,shft keys

- keypress event don't fire when any of them is pressed
- Their properties is set to true only on onkeydown event

event.button value	Description
1	Left Mouse Button
2	Right Mouse Button
4	Middle Mouse Button

Example!

Event Object Properties

Event Object Property	Description
eventPhase	Indicates which phase of the event is evaluated
cancelBubble (deprecated)	Can cancel an event bubble

event.eventPhase value	Constant Property	Description
0	Event.NONE	No event is being processed at this time.
1	Event.CAPTURING_PHASE	The event is being propagated through the target's ancestor objects
2	Event.AT_TARGET	The event has arrived at target
3	Event.BUBBLING_PHASE	The event is propagating back up through the target's ancestors in reverse order

Example!

Event Object Methods

Methods	Description
event.stopPropagation()	Disables the propagation of the current event in the DOM hierarchy. (IE8 = cancelBubble)
event.stopImmediatePropagation()	prevents other listeners are attached to the same element for the same event fr om being called, no remaining listeners will be called.
event.preventDefault()	To cancel the event if it is cancelable, meaning that any default action normally taken by the implementation as a result of the event will not occur. (IE8 = returnValue)
event.composedPath()	Returns the event's path starting with the target object

Other Useful Methods

Methods	Description
elem.addEventListener()	Registers an event handler function for the specified event on the current object.
elem.removeEventListener()	method to remove an event listener that has been registered with the addEventListener method.
elem.dispatchEvent()	Initializes an event object created by the Event Constructor

Synthetic Events

- To create custom event use Event constructor

  var myEvent= new Event(p1,p2)

  - p1: the name of the custom event type
  - p2: an object with the following Optional properties with false as default value
    - bubbles: indicating whether the event bubbles.
    - cancelable: indicating whether the event can be canceled.
    - ~~composed: indicating whether the event will trigger listeners outside of a shadow root.~~

- To fire the event programmatically use dispatchEvent() on a specific element

  elem.dispatchEvent(myEvent)

Synthetic Events

- To create custom event use CustomEvent constructor

  var evt= new CustomEvent(p1,p2)

  - p1: the name of the custom event type
  - p2: is object with details property to add more data to the event object

- To fire the event programmatically use dispatchEvent() on the element registering the event

  elem.dispatchEvent(evt)

Assignments